INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

SURFACE HARDENING: FLAME AND INDUCTION (FOR FERROUS ALLOYS)

REACTIVATED AFTER 18 MAY 2026 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the surface hardening of ferrous alloys by gas flame or electromagnetic induction heating followed by a quenching or relatively rapid cooling operation (see 6.1 and 3.3 and 3.5).

1.2 Classification. Surface hardening is classified according to the following heating methods:

Method I - Flame hardening

Method II - Induction hardening

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM E1444/E1444M	-	Particle Testing, Magnetic
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies of these documents are available online at https://www.astm.org).

SAE INTERNATIONAL

AMS 2745	-	Induction Hardening of Steel Parts

(Copies of these documents are available online at https://www.sae.org/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. Unless otherwise specified in the contract or purchase order, prior to the surface hardening of any parts, the contractor shall submit sample parts or approved control specimens processed by the procedure to be used in production (see 6.2). A processing procedure for each individual part shall be prepared including the factors shown in table I for the applicable method of processing. The submitted procedures shall be considered acceptable if the sample parts or control specimens meet all the requirements of the drawing and this specification.

3.2 Procedural changes. Unless otherwise specified by the procuring activity, any major change in any one of the procedure factors listed in table I, without the approval of the procuring activity, shall require a resubmittal of sample parts reflecting the new processing procedure as well as a record of the new processing procedure.

Table I. Procedure factors.

Factor	Method I	Method II [1/]
a	Size, weight, shape of part	
b	Composition of material	
c	Condition, mechanical properties previous heat treatment (see 3.3 and 3.5)	
d	Required hardness range	
e	Required depth of hardness	
f	Method of supporting and holding or clamping part	
g	Method of moving part or flame	Method of moving work or induction head (heating coil)
h	Fuel gas used	Induction frequency of heating current (Hz)
i	Pressures and regulator and gage settings	Energy requirement (kVA)
j	Type(s) and size(s) of heating tip(s)	Induction head (heating coil) design
k	Flame characteristics and flame inner-cone-to-work distance	Distance between induction head(s) and work (flux air gap)
l	Rate of travel of heating tip(s) or work	Rate of travel of work or induction head(s)
m	Quenching method and medium	
n	Coolant temperature, volume, temperature control arrangement (if any)	
o	Gas (fuel) ana oxygen consumption per part or per unit of surface treated	Energy (kWh) requirement per part er per unit of surface treated
p	Heating time required per part surface hardened	
q	Time between end of heating cycle and start of quenching cycle	
r	Temperature and time of stress relieving	
s	Design of work holding fixture	

[1/] SAE-AMS 2745, “Induction Hardening of Steel Parts”, may be used for method II when greater precision is required.

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3.3 Surface preparation. Surfaces to be hardened shall be clean and free of any decarburization to assure proper response to the surface hardening operation.

3.4 Hardness. The parts shall meet the hardness requirements of the applicable drawing after the surface hardening operation. If hardness values are not specified, the minimum hardness number shall be, after tempering, (72HRA or 43HRC).

3.4.1 Depth of hardening. The depth of hardening is the “effective case depth". Unless otherwise specified, the hardness value to measure the effective case depth should be per table II:

Table II. Hardening depth.

Material Carbon Content (%)	Effective Case Depth Hardness (R_C)
0.28 to 0.32	35
0.33 to 0.43	40
0.43 to 0.52	45
0.53 and over	50

3.4.2 Area of hardening. When only localized areas are to be hardened, the areas shall be so designated on the applicable drawing by dimensionally indicating the basic area to be hardened as well as the allowable overlap.

3.4.3 Core hardness. Core hardness shall be as specified on the applicable drawing. Areas that are not surface hardened shall have a surface hardness equal to the core hardness requirement.

3.5 Prior heat treatment. All parts shall be properly heat treated to the specified core hardness prior to surface hardening.

3.6 Stress relief. Stress relief shall be as specified on the applicable drawings or as authorized by the procuring activity. However, if no stress relief is specified, all surface hardened parts shall, after hardening, be tempered at a minimum tempering temperature of 400°F. for at least one hour per each inch of part thickness. In order to avoid temper embrittlement, the contractor should exercise proper discretion in the selection of a tempering temperature for the steel involved.

3.7 Workmanship.

3.7.1 Surface defects. Surface hardened parts shall be excessive scaling, blisters or other detrimental defects Unless otherwise specified in the contract or purchase order, preproduction sample parts shall be subject to magnetic particle inspection for workmanship defects prior to approval of the surface hardening process.

3.7.2 Warpage. The surface hardening process shall cause no warpage or dimensional changes, net in conformance with the applicable drawings. Straightening or other corrective measures shall not be permitted unless approved by the procuring activity.

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4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).
b. Acceptance inspection (see 4.7).

4.3 Preproduction inspection. Preproduction inspection, except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the production acceptance inspection.

4.4 Lot for acceptance inspection. A lot shall consist of all parts of the same drawing or part number, surface hardened by the same process, and presented for inspection at one time.

4.5 Sampling.

4.5.1 Preproduction. The number of samples shall be specified by the procuring activity (see 6.2).

4.5.2 Acceptance.

4.5.2.1 Hardness. Unless otherwise specified by the procuring activity, sampling for hardness shall follow MIL-STD-1916 and ANSI/ASQ Z1.4 guidance. The Acceptable Quality - Level (AQL) shall be 4.0 percent defective.

4.5.2.1 Depth of hardening. Sampling for depth of hardening shall be as specified by the procuring activity (see 6.2).

4.6 Examination.

4.6.1 Visual. The samples shall be examined in accordance with the sampling procedures specified by the procuring activity for compliance with the surface requirements of 3.7.1 (see 6.2).

4.6.2 Warpage. The samples selected for visual examination in 4.5.1 shall also be measured for compliance with the warpage requirements of 3.7.2.

4.7 Tests.

4.7.1 Test Specimens.

4.7.1.1 Hardness. Specimens shall be prepared in accordance with ASTM E18 as applicable.

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4.7.1.2 Depth of hardening. The sample parts or control specimens shall be sectioned perpendicular to the hardened surface, and the resulting cut face shall be metallographically polished.

4.7.2 Test Procedure.

4.7.2.1 Hardness. Hardness tests shall be conducted.in accordance with ASTM E18 as applicable.

4.7.2.2 Depth of hardening. Depth of hardness shall be determined by etching the polished face and measuring by conducting a hardness traverse through the hardened zone perpendicular to the surface.

4.7.2.3 Magnetic particle inspection. Unless otherwise specified in the contract or purchase order, magnetic particle inspection in accordance with ASTM E1444/E1444M shall be used to check preproduction samples for surface defects.

4.8 Rejection and retests.

4.8.1 Preproduction. The failure of preproduction samples to comply with any of the requirements of this specification shall be cause for rejection of the surface hardening process. Retesting shall not be conducted until it is demonstrated, to the satisfaction of the procuring activity, that the process has been corrected.

4.8.2 Acceptance. Unless otherwise specified in the contract or order, rejection and retest shall be conducted in accordance with the general section of Fed. Test Method Std. No. 151 (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The procedures covered in this specification are intended for use in surface hardening ferrous parts, such as, shafts, pinions, bearing components, worms or spur or bevel gears, etc.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Method of surface hardening, if only one method is allowable (see 1.2).
c. Number and destination of preproduction samples, if required (see 3.1 and 4.5.1).
d. Number of acceptance samples (see 4.5.2.1, 4.5.2.2, and 4.6.1).

6.2.1 Alternate sampling. When alternate sampling is indicated, or requested, the procuring activity should follow MIL-STD-1916 and ANSI/ASQ Z1.4 guidance.

6.3 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table III.

TABLE III. Conversion factors. 1/

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

1/ Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.4 Method II enhancements. For additional guidance and precision for method II, users should refer to SAE-AMS 2745 - Induction Hardening of Steel Parts.

6.5 Subject term (key word) listing.

Gears	Heat Treat	Steel
Hardness	Mechanical	Wear

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - SH
Air Force - 20

Preparing activity:
Army - MR

(Project MFFP-2026-004)

Review activities:
Army - AR, AT, MI

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